DOMiNANCE CVS EXAM PREPARATION

MOCK 03 | ANSWER AND EXPLANATION GUIDE

Answers and explanations from the source-mapped lecture question bank

KEY: 1B 2A 3A 4A 5A 6D 7A 8B 9C 10C 11B 12A 13A 14B 15B 16C 17D 18D 19C 20B 21C 22D 23C 24B 25D 26D 27C 28B 29A 30D 31D 32D 33C 34C 35C 36A 37D 38A 39B 40B

01. B - At the beginning of the QRS complex.

WHY: The PR interval measures from the onset of atrial depolarisation (P-wave onset) to the onset of ventricular depolarisation (QRS onset).

NOT A: The start of T occurs after ventricular depolarisation.

NOT C: Including all of QRS would extend beyond the PR interval.

NOT D: The end of T is used to mark the end of the QT interval.

SOURCE: lecture-7:Q19; 6-7

02. A - Renin-angiotensin effects and renal salt retention.

WHY: The rapid baroreflex stabilizes pressure within seconds, while the renin-angiotensin-aldosterone system supports longer-term pressure and volume.

NOT B: Aldosterone supports sodium retention rather than its suppression.

NOT C: Baroreflexes remain available during sustained standing.

NOT D: Loss of arteriolar responsiveness would worsen orthostatic control.

SOURCE: circulation-3:Q9; Circulation 3, slide 4; Chapter 19 3

03. A - Cortical signals acting through the hypothalamus on autonomic centres.

WHY: Emotions such as anger begin in the cortex. The cortex drives the hypothalamus, which excites the vasomotor centre, raising sympathetic output so heart rate and blood pressure rise.

NOT B: Renal volume control is too slow and not the primary emotional route.

NOT C: The CSF pathways do not directly account for this autonomic response.

NOT D: Platelet serotonin is a local vascular constrictor, not the main emotional signal.

SOURCE: circulation-2:Q57; 30, 33

04. A - Retinal perfusion falls as blood shifts toward the feet.

WHY: Positive g causes leg pooling and reduced pressure at eye level, so retinal perfusion can fail before cerebral consciousness.

NOT B: Pressure at eye level decreases during positive g.

NOT C: The mechanism is reduced eye-level arterial perfusion, not an inevitable ICP excess.

NOT D: Headward blood movement occurs during negative g.

SOURCE: circulation-3:Q16; Circulation 3, slide 9; Chapter 19 6

05. A - All valves closed, ventricular pressure rises, volume fixed.

WHY: Both ventricles generate rising pressure with all inlet and outlet valves closed, so neither receives nor ejects blood.

NOT B: Open valves permit volume exchange.

NOT C: AV valves do not remain open once ventricular pressure rises above atrial.

NOT D: Falling pressure describes isovolumetric relaxation, not contraction.

SOURCE: circulation-1:Q19; 14-15

06. D - Reduced plasma oncotic pressure promotes oedema.

WHY: Proteinuria reduces circulating albumin and its inward oncotic force, favouring movement of fluid into interstitium.

NOT A: Viscosity is not the principal mechanism of protein-loss oedema.

NOT B: Capillary permeability elsewhere need not decrease.

NOT C: Albumin loss lowers oncotic pressure.

SOURCE: circulation-1:Q88; 93

07. A - Ventricular contraction compresses intramural vessels.

WHY: High left ventricular wall tension during systole squeezes intramyocardial coronary vessels; relaxation permits better flow.

NOT B: Coronary ostia remain open through the cardiac cycle.

NOT C: Aortic pressure persists during diastole and drives coronary flow.

NOT D: Coronary flow increases during relaxation rather than disappearing.

SOURCE: lecture-6:Q22; 27; Chapter 23 3

08. B - Increased cAMP/PKA signalling -> greater calcium availability.

WHY: Beta-1 signalling raises cAMP and PKA activity, enhancing calcium entry and cycling for stronger contraction.

NOT A: Blocking release channels would reduce systolic calcium release.

NOT C: Vagal slowing is not the main ventricular inotropic pathway.

NOT D: Beta-1 stimulation raises, not lowers, cAMP.

SOURCE: circulation-1:Q10; 46-47; Chapter 2 7

09. C - Deep vein compression with one-way valves.

WHY: Calf contractions compress deep veins, while competent venous valves favour upward flow toward the heart instead of backflow.

NOT A: Venous return requires directed pressure gradients, not dilation alone.

NOT B: Capillaries do not function as contractile muscle pumps.

NOT D: Arteries do not rely on venous-style one-way valves.

SOURCE: circulation-3:Q6; Circulation 3, slide 6; Chapter 19 3

10. C - Direct vasomotor-centre excitation greatly increases sympathetic outflow.

WHY: Hypoxic, hypercapnic medullary tissue drives a strong sympathetic pressor response to restore cerebral blood flow.

NOT A: Inhibition would tend to lower vascular tone and pressure.

NOT B: Vagal discharge slows the heart and is not the primary pressor mechanism.

NOT D: ANP favours dilation and volume loss.

SOURCE: circulation-2:Q54; 31-32, 35-36

11. B - Several leads are calculated from combinations of the same electrode voltages.

WHY: A lead is an electrical viewpoint calculated from one or more electrode potentials, so ten physical contacts can generate twelve different standard recordings.

NOT A: Routine twelve-lead ECG does not require intracardiac electrodes.

NOT C: All twelve are electrical leads; pressure sensors do not generate extra leads.

NOT D: The limb contacts remain in place during a standard recording.

SOURCE: lecture-7:Q14; 3-5; Chapter 6 3

12. A - It shortens the heart-to-brain hydrostatic column.

WHY: A more horizontal posture decreases the vertical distance over which acceleration reduces pressure toward the head and brain.

NOT B: Increasing the vertical distance would worsen the gradient.

NOT C: Enhanced ankle pooling would reduce central venous return.

NOT D: Greater dependent filtration would not protect brain perfusion.

SOURCE: circulation-3:Q17; Circulation 3, slide 11; Chapter 19 6

13. A - A small or biphasic ventricular deflection.

WHY: Only the component of the electrical vector along the lead axis contributes strongly to the deflection; a near-perpendicular vector has a small projection.

NOT B: A strongly downward deflection needs a substantial vector component away from the positive pole.

NOT C: A strongly upright deflection needs a substantial vector component towards the positive pole.

NOT D: Vector orientation changes amplitude and polarity, not necessarily conduction duration.

SOURCE: lecture-7:Q3; 3-5; Chapter 6 2

14. B - 4 L/min.

WHY: Flow equals pressure difference divided by resistance: 80/20 = 4 L/min. Increasing resistance with the same pressure difference would lower flow.

NOT A: Two L/min would correspond to resistance of 40.

NOT C: Eight L/min would correspond to resistance of 10.

NOT D: Three L/min does not fit the given ratio.

SOURCE: circulation-1:Q50; 61, 67-70

15. B - Intermittent cannon a waves.

WHY: Unsynchronised atrial contraction against a shut AV valve can produce large intermittent cannon a waves.

NOT A: Giant v waves primarily suggest tricuspid regurgitation.

NOT C: Atrial contractions still cause pressure variations.

NOT D: Venous pulsations can persist despite AV block.

SOURCE: circulation-1:Q72; 26; Chapter 11 7

16. C - It falls by about 5 mmHg.

WHY: The arterial-venous difference changes by -30 minus -25, or -5 mmHg. Falling venous and intracranial pressures therefore help protect cerebral perfusion.

NOT A: That ignores the fall in cerebral venous pressure.

NOT B: The difference becomes slightly smaller, not larger.

NOT D: The changes cannot be added as two rises.

SOURCE: circulation-3:Q7; Circulation 3, slide 5; Chapter 19 3

17. D - Antidromic impulses in branches of a sensory axon.

WHY: The flare is an axon reflex: sensory-nerve branches release vasodilator neuropeptides in surrounding skin without a spinal reflex.

NOT A: Lymphatic filtration does not produce this arteriolar flare.

NOT B: The surrounding redness is functional vasodilation, not thrombosis.

NOT C: A central synaptic reflex is not needed for the axon reflex.

SOURCE: lecture-6:Q51; 40

18. D - Raised capillary pressure increases outward filtration.

WHY: The increased hydrostatic pressure in dependent leg capillaries pushes more fluid into the interstitium and promotes oedema.

NOT A: A lower capillary pressure would oppose outward filtration.

NOT B: Standing promotes dependent filtration; it does not stop capillary exchange.

NOT C: Plasma oncotic pressure ordinarily favours fluid retention within blood.

SOURCE: circulation-3:Q4; Circulation 3, slide 3; Chapter 19 2

19. C - Rapid passive ventricular filling.

WHY: S3 occurs during rapid early-diastolic ventricular filling, as blood entering the chamber causes vibrations.

NOT A: Atrial depolarisation precedes the atrial contribution later in diastole.

NOT B: Isovolumetric contraction occurs near S1.

NOT D: Aortic valve opening occurs during ventricular systole.

SOURCE: circulation-1:Q29; 28

20. B - A possible late ventricular repolarisation phenomenon.

WHY: The lecture describes a U wave, when seen after T, as the final remnants of ventricular repolarisation. Its precise physiological origin is uncertain.

NOT A: Atrial depolarisation is the P wave preceding QRS.

NOT C: Rapid ventricular depolarisation is represented by QRS, not U.

NOT D: The ventricular plateau is associated with the ST segment.

SOURCE: lecture-7:Q29; 10

21. C - To adjust body heat gain and loss.

WHY: Cutaneous vessels dilate to increase heat loss and constrict to conserve heat; flow can far exceed nutritional needs.

NOT A: Skin metabolic demand is relatively modest compared with variation in perfusion.

NOT B: Heart pacemaking is not the main thermoregulatory role of skin circulation.

NOT D: Pulmonary-systemic pressure equality is unrelated to thermoregulation.

SOURCE: lecture-6:Q46; 39-41

22. D - The ventricle must generate higher pressure for ejection.

WHY: Higher arterial pressure increases ventricular pressure work and wall stress, requiring more oxygen for force generation.

NOT A: Higher afterload generally increases wall stress.

NOT B: The myocardium continues to contract against the higher load.

NOT C: Coronary oxygen extraction remains substantial.

SOURCE: circulation-1:Q48; 54; Chapter 8 7

23. C - Its small electrical signal is obscured by the ventricular QRS complex.

WHY: Atrial repolarisation occurs around the time of ventricular depolarisation; the much larger QRS electrical signal masks it on the surface trace.

NOT A: Repolarisation does generate voltage differences, although the signal is small.

NOT B: Atrial repolarisation does not need to await the beginning of ventricular diastole.

NOT D: That statement does not explain why the wave is not separately visible.

SOURCE: lecture-7:Q25; Chapter 6 4 (approved-chapter explanation)

24. B - It lasts 0.14 s and is longer than the stated usual range.

WHY: At 25 mm/s, 3.5 small squares equal 3.5 x 0.04 = 0.14 s, above the lecture's usual QRS duration of 0.06-0.10 s.

NOT A: 0.10 s corresponds to 2.5 small squares at the standard speed.

NOT C: 0.07 s incorrectly halves the 0.14-s measured duration.

NOT D: 0.35 s does not follow from the standard 0.04-s small-square duration.

SOURCE: lecture-7:Q39; 8, 13

25. D - Constriction of systemic arterioles and veins.

WHY: Sympatholytic drugs blunt reflex arteriolar and venous constriction, so the circulation cannot adequately oppose blood pooling and falling arterial pressure on standing.

NOT A: Walking uses a mechanical muscle pump, not this autonomic drug pathway.

NOT B: Venous valve directionality is mechanical and not directly abolished by sympatholytic drugs.

NOT C: Cerebral metabolic vasodilatation is not directly a sympathetic vasoconstrictor response.

SOURCE: circulation-3:Q15; Circulation 3, slide 8; Chapter 19 5

26. D - Compression of brain vessels and reduced driving pressure.

WHY: Rising intracranial pressure compresses intracranial vessels and lowers cerebral perfusion pressure, risking medullary ischaemia.

NOT A: Raised intracranial pressure does not require systemic blood loss.

NOT B: Dilation cannot prevent the loss of cerebral driving pressure as intracranial pressure rises.

NOT C: Perfusion initially becomes harder as intracranial pressure rises.

SOURCE: circulation-2:Q53; 31-32, 35-36

27. C - Aortic pressure exceeds ventricular pressure.

WHY: When ventricular pressure falls below aortic pressure, brief reversal tendency closes the aortic valve and starts isovolumetric relaxation.

NOT A: Venous pressure does not directly determine aortic valve closure.

NOT B: Atrial pressure governs mitral opening, not aortic closure.

NOT D: Ventricular pressure exceeding aortic pressure opens the valve.

SOURCE: circulation-1:Q23; 17

28. B - Red line, flare and wheal.

WHY: The classical triple response consists of local redness, a spreading flare and an oedematous wheal.

NOT A: A white line is a separate lighter-stroke reaction.

NOT C: Wheal formation requires local fluid leak rather than none.

NOT D: Thrombosis and scar are not normal features.

SOURCE: lecture-6:Q52; 40

29. A - Their operating range resets around the sustained level.

WHY: When pressure stays high for days, the baroreceptors reset and treat the new level as normal, so they defend it instead of correcting it. Long-term pressure is set mainly by the kidneys' control of salt and water.

NOT B: Unbounded sensitivity would strongly oppose the persistent rise.

NOT C: Sensory fibres do not become motor fibres.

NOT D: Resetting does not require loss of every afferent signal.

SOURCE: circulation-2:Q44; 18, 24-27

30. D - A brief local vasoconstrictor response.

WHY: The lecture describes a transient white reaction due to constriction; it is distinct from the classical red-flare-wheal response.

NOT A: Wheal is raised local oedema from permeability.

NOT B: The red line is a separate vasodilator response.

NOT C: Flare is surrounding redness from arteriolar dilation.

SOURCE: lecture-6:Q49; 39-40

31. D - Overlapping endothelial flaps admit interstitial fluid.

WHY: Initial lymphatic endothelial flaps allow interstitial fluid and proteins to enter when surrounding pressure exceeds lymphatic pressure.

NOT A: Initial lymphatic capillaries lack a thick continuous smooth-muscle coat.

NOT B: There is no single central lymphatic heart.

NOT C: Initial lymphatics are relatively permeable to interstitial proteins.

SOURCE: circulation-1:Q93; 60; Chapter 15 2

32. D - Other conducting tissues retain latent automaticity.

WHY: Subsidiary pacemakers in the AV junction or Purkinje system can spontaneously depolarise, usually at slower rates.

NOT A: Autonomic nerves modify pacemaking but do not directly activate every myocyte.

NOT B: Coordinated contraction still requires electrical activation.

NOT C: Valves are passive mechanical structures.

SOURCE: circulation-1:Q13; 7; Chapter 3 4

33. C - The arterial baroreflex reacts to the sympathetic pressure rise.

WHY: Medullary ischaemia first raises sympathetic pressure; arterial baroreceptors then increase vagal drive and slow the heart.

NOT A: beta1 stimulation would usually accelerate, not slow, the heart.

NOT B: Cardiac autonomic signalling remains central to the response.

NOT D: Atrial volume receptors respond chiefly to filling, not this sequence.

SOURCE: circulation-2:Q55; 32-36

34. C - Hypoxia together with accumulating carbon dioxide.

WHY: Poor medullary perfusion produces hypoxia and hypercapnia, which strongly stimulate the vasomotor centre to raise pressure.

NOT A: High oxygen and low CO2 do not represent medullary ischaemia.

NOT B: Low acidity and abundant oxygen oppose the described stimulus.

NOT D: Adequate oxygen delivery would not produce the ischaemic response.

SOURCE: circulation-2:Q52; 30-32

35. C - Pulmonary artery -> lung capillaries -> pulmonary veins.

WHY: The right ventricle pumps blood through pulmonary arteries to lung capillaries; pulmonary veins return it to the left atrium.

NOT A: The aorta receives systemic output from the left ventricle.

NOT B: That is mainly the systemic circulation after left ventricular ejection.

NOT D: Pulmonary veins carry blood into the left atrium rather than away from the right ventricle.

SOURCE: circulation-1:Q1; 2-6

36. A - NTS activation increases depressor and vagal signalling.

WHY: More stretch -> faster baroreceptor firing -> more input to the NTS. The NTS excites the depressor area (which inhibits the pressor area) and the vagal neurons, so sympathetic output falls and vagal output rises.

NOT B: The heart remains under autonomic regulation during the reflex.

NOT C: The reflex suppresses rather than increases sympathetic pressor drive.

NOT D: Increased wall stretch increases, rather than decreases, afferent traffic.

SOURCE: circulation-2:Q40; 22-24

37. D - Venoconstriction shifts stored blood centrally.

WHY: Sympathetic venoconstriction reduces the capacity of the venous reservoir and shifts stored blood toward the heart, supporting venous return and end-diastolic filling.

NOT A: Systemic arterioles have no valves that close with sympathetic stimulation.

NOT B: Venodilation generally increases peripheral capacitance and pooling.

NOT C: Venoconstriction redistributes existing blood rather than rapidly generating plasma.

SOURCE: circulation-1:Q73; 73-75, 97-98; Chapter 12 2

38. A - Prolonged ischaemia can produce myocyte necrosis.

WHY: Prolonged severe ischaemia can cause irreversible myocardial cell death; transient angina does not necessarily cause necrosis.

NOT B: Infarction can occur across a range of heart rates.

NOT C: Transient angina ordinarily reflects reversible myocardial ischaemia.

NOT D: Stable angina may result from fixed stenosis without rupture.

SOURCE: lecture-6:Q28; 29

39. B - The elastic recoil of large arteries sustains diastolic flow.

WHY: Elastic arteries stretch during systole and recoil during diastole, smoothing pulsatile ejection into continuous downstream flow.

NOT A: Venous valves prevent backflow but do not create aortic pressure.

NOT C: Respiratory movement mainly assists venous return.

NOT D: Capillaries lack the pumping muscle required to drive systemic flow.

SOURCE: circulation-1:Q49; 55-57

40. B - Restricted coronary flow reserve cannot meet increased demand.

WHY: A narrowed vessel may deliver adequate resting flow but cannot dilate enough to meet exercise demand, causing reversible ischaemia.

NOT A: Plaque does not disappear with exertion.

NOT C: Exercise raises myocardial work and oxygen requirements.

NOT D: Resting coronary flow may be sufficient despite reduced reserve.

SOURCE: lecture-6:Q27; 29; Chapter 23 6